

Basic Mathematical Definitions *and* Using Python for Cryptography

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$$g^{a^b} = g^{b^a} = g^{ab}$$

```
1 def exp():
2     g = 9 # generator
3     a = 13 # a's secret key (aka private key)
4     a_public = pow(g, a) # g^a
5     b = 42 # b's secret key
6     b_public = pow(g, b) # g^b
7     # @a shares their public key `a_public` with @b
8     ss_b = pow(a_public, b_secret)
9     # @b shares their public key `b_public` with @a
10    ss_a = pow(b_public, a_secret)
11    assert ss_a == ss_b
```

$$g^{a^b} = g^{b^a} = g^{ab}$$

```
1 def mod_exp():
2     p = 3329 # prime modulus
3     g = 9 # generator
4     a_secret = 13
5     a_public = pow(g, a_secret, p) # g^a_secret mod p
6     b_secret = 42
7     b_public = pow(g, b_secret, p) # g^b_secret mod p
8     ss_alice = pow(b_public, a_secret, p)
9     ss_bob = pow(a_public, b_secret, p)
10    assert ss_alice == ss_bob
```

$$g^{a^b} \equiv g^{b^a} \equiv g^{ab} \pmod{p}$$

Given

g , p , and $g^a \pmod{p}$,

find a

Function

Domain

Codomain

Range

Function is a mapping from a *domain* to a *codomain*.

Range is the subset of the codomain that is *actually* mapped to by the function.

Image of a function f is the set of all outputs of f .

Preimage of a function f is the set of all inputs that map to a given output of f .

Given a function f and an input x , a **second preimage** of x is an input $x' \neq x$ such that $f(x) = f(x')$.

A **one-way function** is a function that is easy to compute but hard to invert.

More formally, a function f is a one-way function if:

- f can be computed in polynomial time.
- For every probabilistic polynomial-time algorithm A , the probability that $A(f(x))$ outputs a preimage of $f(x)$ is negligible.

A **hash function** is a function that maps data of arbitrary size to a fixed-size value.

A **cryptographic hash function** is a hash function that has the following properties:

- It is easy to compute the hash value for any given input.
- It is infeasible to generate a valid input that has a given hash value.
- It is infeasible to find two different inputs that produce the same hash value.

A **cryptographically secure pseudorandom number generator** (CSPRNG) is a pseudorandom number generator that has the following properties:

- It is computationally indistinguishable from a true random number generator.
- It is resistant to state compromise extensions, meaning that even if an attacker learns the internal state of the generator at some point, they cannot predict future outputs.